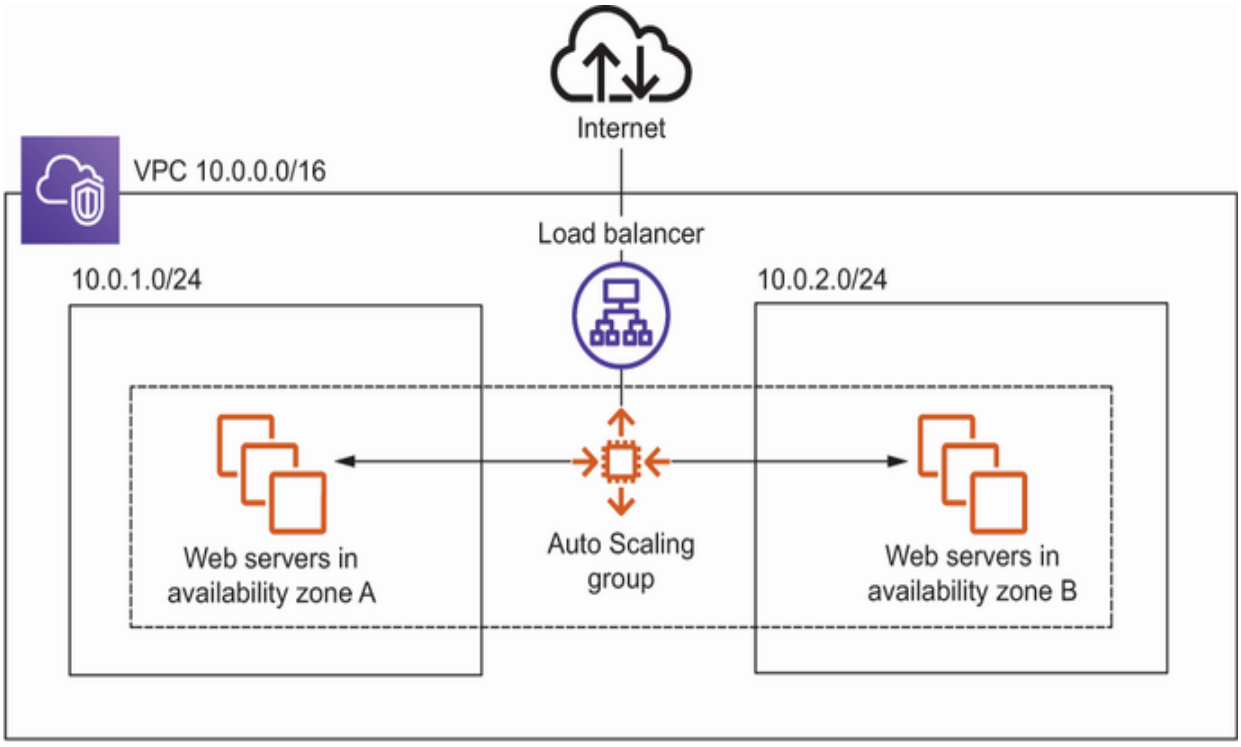


Figure 16.3 Fault-tolerant EC2 instances with an Auto Scaling group and an Elastic Load Balancer

Take a second look at figure 16.3 and see what parts are redundant:

- *Availability zones (AZs)*—Two are used. If one AZ suffers from an outage, we still have instances running in the other AZ.
- *Subnets*—A subnet is tightly coupled to an AZ. Therefore, we need one subnet in each AZ.
- *EC2 instances*—Two subnets with one or more EC2 instances lead to redundancy among availability zones.
- *Load Balancer*—The load balancer spans multiple subnets and, therefore, multiple availability zones.

Next, figure 16.4 shows a fault-tolerant system built with EC2 that uses the power of redundancy and asynchronous decoupling to